Predicting Food Spoilage with Machine Learning

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MOTIVATION

One in six Americans contract a foodborne illness every year[4].

Existing mechanistic methods require per-condition calibration.

Existing deep learning methods lack biological grounding.

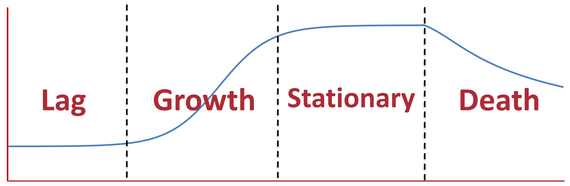
Predictions are difficult to generate and interpret by non-experts.

OBJECTIVES

1. Build a model that predicts spoilage using time-series data and environmental conditions.
2. Implement a chatbot that returns plain-English explanations of shelf-life predictions.

BACKGROUND

BACTERIAL POPULATION GROWTH PHASES

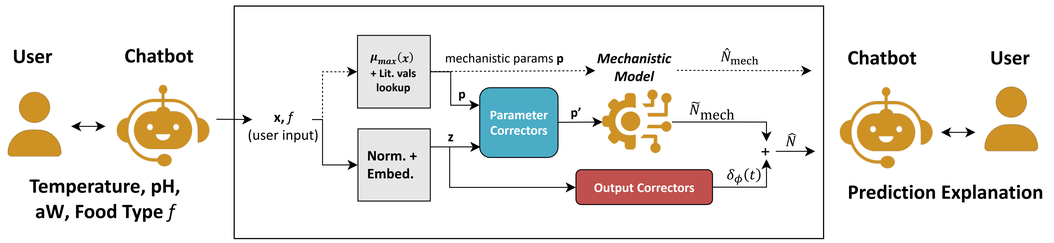


Growth phase depends on water activity (aW), pH, temperature and bacteria present.

Initial bacteria count is unknown at purchase – a key reason spoilage risk differs for same food and same fridge.

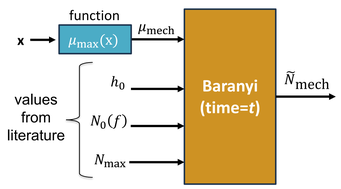
METHODOLOGY

ComBase Preprocessing: ComBase[1] is a public database of bacterial growth curves submitted by independent scientists across many labs, capturing organism, food type, temperature, pH and water activity. Curves are filtered, cleaned, and mapped to standardized food names.

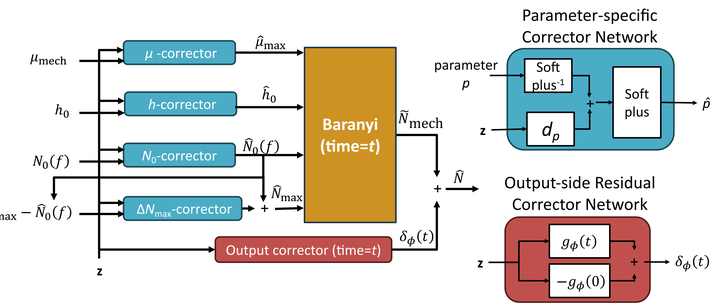


ARCHITECTURE

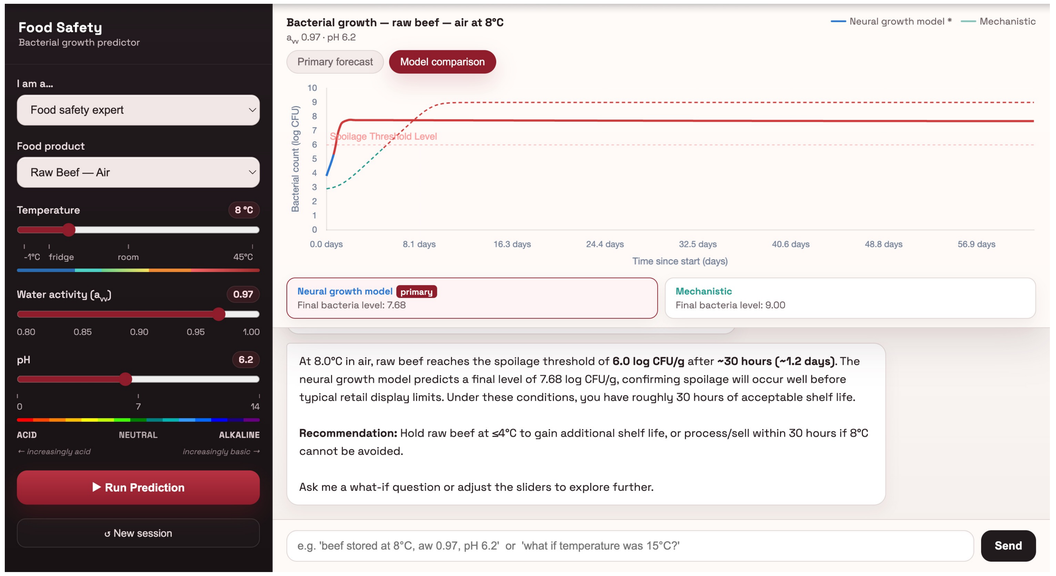
Baseline (mechanistic)[2,3]



Neural Reparameterizer and Corrector Model



CHATBOT



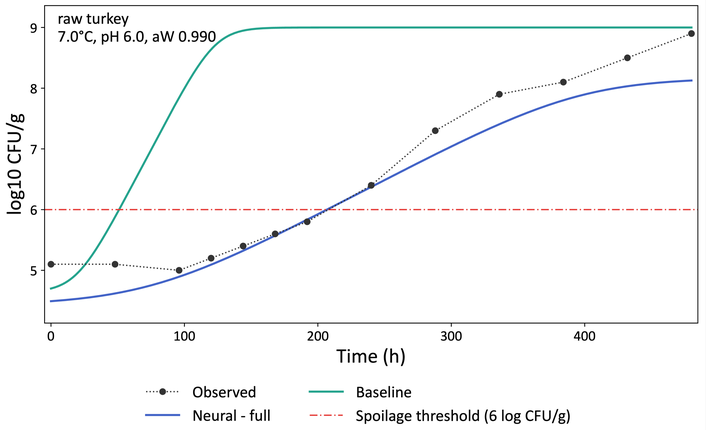
RESULTS

Benchmark: Shared, stratified 5-fold cross-validation

Training Population: 245 curves | 2,968 observations

Model	Curve-fit	Shelf-Life Detection			
	RMSE	Timing MAE (days)	Censored Score (days)	Miss Rate	False-Alarm Rate
Baseline	2.89	4.88	4.72	0.35	0.57
Neural *(no output corr.)*	1.40	3.11	3.38	0.15	0.32
Neural *(no param corr.)*	1.70	3.04	3.14	0.10	0.68
Neural *(full)*	1.41	3.10	3.37	0.14	0.32

Relative to the mechanistic baseline, our full neural model reduced the *false alarm rate* by **43.9%** and the *censored score* by **28.6%**.



CONCLUSION

A mechanistic-neural hybrid model that performs both parameter- and output-space corrections improves both the fit and generalization of the data.

Our chatbot serves as a practical tool for intuitively generating predictions and explanations.

FUTURE WORK

Expanded data coverage, uncertainty estimation, and food-to-spoilage-flora representation.

[1]Baranyi and Tamplin, "ComBase: A Common Database on Microbial Responses to Food Environments," *Journal of Food Protection*, 2004,
[2]Baranyi and Roberts, "A Dynamic Approach to Predicting Bacterial Growth in Food," *International Journal of Food Microbiology*, 1994.
[3]Leroi, Fall, Pilet, Chevalier, and Baron, "Influence of Temperature, pH and NaCl Concentration on the Maximal Growth Rate of Brochothrix Thermosphacta and a Bioprotective Bacteria Lactococcus Piscium CNCM I-4031," *Food Microbiology*, 2012.
[4]United States Food and Drug Administration, "People at Risk of Foodborne Illness," 2020 [Online].

Our work is supported by the National Science Foundation, Grant 2349370 and the WPI Summer Training in Arts & Sciences Research (STAR) Fellowship.
We would also like to thank the USDA ComBase Maintainers and the Karsh STEM Scholars Program.